TN SKILLS

EduGenie: Google Gemini Powered Learning Assistant

Brainstorming & Ideation Phase

- The brainstorming phase identifies the learning problems students face and converts those problems into useful AI-based features.
- Students may find it difficult to understand complex topics, obtain immediate answers, revise large amounts of content, prepare questions for exams and maintain a consistent study routine.
- EduGenie acts as a virtual learning companion. Students can enter a question or topic and receive an explanation generated by Gemini. Additional learning tools can transform the same topic into summaries, flashcards, practice questions and quizzes.
- AI chat assistant; topic explanation; text summarization; quiz generation; question generation; study recommendations; learning history; difficulty-level selection; and responsive web access.
- A clear project concept with a defined target audience, problem statement, feature list and initial scope.

Requirement Analysis Phase

- The application should allow students to enter questions, send prompts to Gemini, display AI responses, request summaries, generate quizzes, and interact with the assistant through a simple interface.
- Users need an easy login or access mechanism, a clear chat interface, readable responses, useful error messages and fast interaction.
- The system requires a Gemini-compatible API integration, prompt templates, safe handling of user input, response processing and appropriate error handling when the AI service is unavailable.
- The application should be responsive, reliable, secure, maintainable, reasonably fast and usable on both desktop and mobile devices.
- A normal development computer, internet connection, VS Code or another IDE, a supported runtime such as Python/Node.js depending on implementation, a database if persistence is required, and access to the Google Gemini API.
- The result of this phase is a Software Requirements Specification containing project scope, users, features, constraints and technical requirements.

Project Design Phase

- The interface can contain a navigation area, chat window, prompt box, send button, quick learning actions, quiz area and response cards.
- The design should use clear typography and responsive layouts.
- If persistence is required, entities may include User, ChatSession, Message, Quiz, QuizQuestion and LearningHistory. Sensitive API credentials should never be stored in client-side code.
- Prompts should define the assistant role, educational objective, expected answer format, learner level and constraints such as clarity and

Project Planning Phase

- The project can be divided into requirement analysis, UI development, backend development, Gemini integration, database integration, testing, documentation and final demonstration.
- Possible roles include project coordinator, frontend developer, backend developer, AI integration developer, tester and documentation/presentation lead. A small project may combine several roles.
- Potential risks include API failures, incorrect AI responses, excessive API usage, security problems and insufficient testing.
- Mitigation includes validation, rate limiting, safe prompt design, error handling and human review of important educational content.
- Project plan, task schedule, responsibility

Project Testing Phase

- Test individual functions such as input validation, prompt creation, response parsing and quiz scoring.
- Verify the connection between frontend, backend and Gemini API. Confirm that requests and responses move correctly through the system.
- Test chat, explanation, summarization, quiz generation, navigation and error messages against the requirements.
- Ask representative students to use the application and identify confusing screens, difficult interactions and unclear responses.
- Measure response time, loading behavior and application stability under repeated requests.
- Test cases, test results, defect reports, fixes and final validation report.

Project Development Phase

- Build the pages and components for the EduGenie interface. Implement the chat experience, prompt input, loading states, response display, quiz interface and responsive design.
- Create API endpoints for chat, summaries, quizzes and other learning actions. Validate requests and return structured responses.
- Configure the Gemini API using a server-side environment variable for the API key. The backend sends carefully constructed prompts and processes the model response before returning it to the client.
- Implement topic explanations, simplified explanations, summaries, question generation, quiz generation and study suggestions.
- Handle empty prompts, invalid requests, API timeouts, service errors and unexpected responses with user-friendly messages.

Project Documentation Phase

- Document the project title, abstract, introduction, problem statement, objectives, existing system, proposed system, requirements, architecture, implementation, testing, results and future enhancements.
- Describe installation steps, environment variables, project structure, API endpoints, database structure and Gemini integration.
- Provide instructions for opening the application, asking questions, generating summaries, taking quizzes and interpreting results.
- Document request formats, response formats, authentication requirements and common errors without exposing secret credentials.
- A complete project report, README, setup guide, testing report and presentation material.
- Explain how developers can update prompts, add new

Project Demonstration Phase

- Prepare the final application, sample questions, quiz examples, system architecture diagram and a short presentation.
- 1) Introduce the problem. 2) Explain EduGenie's objective. 3) Show the home page. 4) Ask a learning question. 5) Demonstrate a simplified explanation. 6) Generate a summary. 7) Generate and attempt a quiz. 8) Explain the Gemini integration and architecture. 9) Show testing results. 10) Present future enhancements.
- The demonstration should show that EduGenie can provide interactive, personalized learning assistance through a simple interface powered by generative AI.
- A student can enter a topic such as 'Explain photosynthesis for a beginner'. EduGenie sends an educational prompt to Gemini and displays a